

Extending Motion Complexity Reduction with Eye Gaze Data to Evaluate Cybersickness

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ABSTRACT

The measurement of cybersickness levels in research are often subjective reports which are prone to reporting bias, creating a need for objective measures. Recent work has proposed motion complexity, using the d95 metric, based on principal component analysis of head movement data, as a potential objective indicator of cybersickness. This study examines the robustness of the d95 metric under constrained motion conditions and examines the utility of including eye tracking data. d95 scores were computed using six degrees of freedom from head movement data (d95.6) and nine degrees of freedom by extending to explore the use eye gaze data (d95.9). While both metrics showed significant relationships with FMS sickness scores when analyzed independently, neither remained significant when analyzed in the combined model where the study's experimental condition was the only significant predictor. Analysis of principal component loadings revealed that translation was the dominant modality in the first principal component, while eye movements were dominant in the second and third principal components. However, translation remained the dominant modality by total variance explained across the three principal components. These results suggest that extending d95 to eye tracking does not improve predictive power and that d95 primarily reflects head movement rather than fine oculomotor behavior.

Index terms: cybersickness, motion complexity, principal component analysis.

1 INTRODUCTION

Cybersickness, a form of visually induced motion sickness (VIMS), an experience resembling motion sickness, may occur while using virtual reality (VR) technology [1]. Symptoms such as nausea, headaches, and eye strain, among others, manifest during VR use for a majority of those who use the technology [1, 15]. With the popularity of VR technology rising and more use cases found every year, more users are exposed to these negative symptoms without clear solutions.

One challenge of cybersickness research is how individualized the experience is. Many different factors can have varying levels of influence on the symptoms and severity experienced, such as sex, age, and gaming experience, along with others that are not as well understood [15, 21]. These individual differences can lead to

different perceptions of the same environment across users and different severity of symptoms.

Cybersickness measurement for VR exposures has primarily relied on subjective surveys completed shortly after the VR experience, namely the Simulator Sickness Questionnaire (SSQ) [10] or Virtual Reality Sickness Questionnaire (VRSQ) [14]. More recently, the Fast Motion Sickness (FMS) scale [11], Misery scale [2], and Discomfort Score (DS) [6] have been used to measure sickness during the VR exposure by asking a single scale-based question at regular intervals. Although these measurements are easily implemented, they are also subjective and therefore prone to reporting bias [22].

Objective measures can be used to measure cybersickness ratings without bias or interpretation differences. Previous research has explored the relationship between cybersickness and physiological measures such as heart rate, heart rate variability, galvanic skin response (GSR), and electroencephalogram (EEG) [4, 19]. Others have investigated eye tracking [17] and head movements [8, 23]. Current barriers of these objective measures include invasiveness and inconvenience while wearing a VR headset, and thus far the literature has not reported a clear predictor of cybersickness based on physiological signals.

One objective measure that has been investigated recently is motion complexity using a VR headset's motion tracking capabilities [3], which follows the loss of complexity hypothesis [16]. This hypothesis states that physiological dynamic complexity can be negatively impacted by diseases or aging. DeGuzman et. al. [3] proposed the d95 complexity score [13], derived from principal component analysis (PCA), to be used to measure motion complexity in cybersickness research and found that the sicker a user was, the less complex their movements were.

This paper has two goals. First, the study examines whether the robustness of the d95 complexity score proposed by DeGuzman et. al. [3] continues to predict cybersickness when behavior in the virtual environment is constrained by the task rather than controlled by the user. Rather than proposing a new objective metric, this work aims to clarify when and why d95 is informative by testing its sensitivity to task constraints and by examining the behavioral sources of variance underlying the metric. Second, the study explores whether extending the d95 score to include eye movement data increases its predictive value compared to using head movement data alone. Eye movement data was chosen to test whether incorporating a second behavioral subsystem (oculomotor) increases the predictive value of motion complexity, as eye data can be easily captured during cybersickness studies and has not yet been analyzed in the motion complexity context.

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2 BACKGROUND

2.1 Subjective and Objective Measures

Subjective measures of cybersickness have been the established method for decades, with the most popular survey being the SSQ [10], which originated from astronaut training in simulators. The SSQ contains sixteen symptom ratings on a four-point Likert scale. The survey is often taken directly before and directly after VR exposure to measure how much a participant's score changed. A more recent survey is the VRSQ that was inspired by the SSQ but focused more on VR usage [14]. These surveys can provide a post-exposure view of cybersickness, but do not provide information on real-time cybersickness ratings. The FMS [11], Misery scale [2], and discomfort score [6] assess cybersickness ratings during exposure, with a 0 – 20, 0 – 10, and 0 – 10 scale respectively. Participants are asked to rate their current level of cybersickness at regular intervals throughout the exposure period. Although well established, these subjective measures are limited by reporting bias [22].

Objective measures are a potential avenue around the limitations of subjective measures. Physiological data has been used to try to identify and predict cybersickness, such as heart rate [9, 12], heart rate variability [9, 12], and GSR [4]. EEG has also been used to investigate the different signals in the brain associated with cybersickness [4]. Eye tracking data has been investigated in the context of cybersickness, such as blink rate [18] and saccade and fixations [17], with differing results. Motion data such as the head tracking information has been used recently to investigate motion complexity [3]. These objective measures are commonly fed into machine learning models as features to help identify current and future experiences of cybersickness [8].

2.2 Motion Complexity and d95

The d95 motion complexity score methodology used in this study is inspired by DeGuzman et. al., [3] who first applied the d95 score to the cybersickness context. The d95, which is the number of principal components needed to explain 95% of variance in principal component analysis, was originally used to evaluate body ownership [13]. DeGuzman et. al. found that by using the d95 score on two experimental datasets, the higher discomfort scores were associated with less flexible and varied movement patterns in VR, reflected by lower d95 scores. This aligns with the loss of complexity hypothesis, which proposes that physiological systems become more constrained under stress or illness.

DeGuzman et al. investigated two formulations of the d95 metric. The first was a 6-DOF model based on the VR headset's physical x, y, z positions and three rotation axes (pitch, roll, yaw). The second was a 9-DOF model that extended this representation by incorporating the x, z, and yaw information of the virtual camera rig. They found that including virtual rig motion increased the sensitivity of d95 to cybersickness in their datasets. Given the promising nature of the results, more studies are needed to see how robust this metric is for detecting cybersickness across task domains.

The present study adopts the same PCA-based framework but applies it to a dataset in which user motion is externally constrained rather than actively controlled. Virtual camera rig motion was not available in this dataset, so we instead extend the d95 metric by incorporating eye gaze data. Although eye movements differ from rig motion in many ways, they represent a distinct behavioral subsystem that can be captured during VR exposure.

3 METHODS

3.1 Research Questions

RQ1. Does d95 continue to predict cybersickness when behavior is constrained by the environment rather than regulated by the user?

RQ2. Does including eye tracking data in the d95 score increase the predictive value beyond the 6-DOF data?

RQ3. How do head and eye movement modalities contribute to the principal components in the d95.9 data?

3.2 Virtual Environment, Procedure, and Participants

The data used in this study is from an attentional cybersickness study performed in Spring 2025 [7]. The study placed participants on a pendulum chair elevated on a pole that swayed according to a formula. The participants had no control over their movement and were tasked with pulling the trigger on their controller when they noticed a particular tree model appeared. Two conditions existed, Center and Whole, with the Center condition spawning trees only in the center of the user's horizontal field of view at a set distance from the user. In the Whole condition, trees could spawn anywhere in the user's horizontal field of view and at varying distances from the user, with adjusted tree scaling to ensure distance spawned was not a confound. The total exposure was 10 minutes, but participants could drop out early if requested.

Throughout the exposure period, participants were asked their FMS every minute. In this study, FMS was on a reduced scale of 0-10, which other work has done for ease of comprehension [5]. The VR headset used was the Varjo XR3, which was used to capture head position and eye tracking data. Prior to the exposure, participants answered general demographic questions and completed the pre-SSQ, and following the exposure completed the post-SSQ.

This study consisted of a total of 82 participants (60 men, 21 women, one non-binary), with an average age of 22.4 years (SD = 3.5). A majority of participants played 0 - 7 hours of video games per week ($n = 54$), while fewer played for 7+ hours per week ($n = 18$) and fewer still for 0 hours per week ($n = 10$).

3.3 Motion Complexity Calculation

Motion complexity was measured using the code provided by the authors of the original DeGuzman et. al. paper [3], modified to interpret the dataset from this study. Similar to the original study, this study uses the physical VR headset's x, y, z position and rotation as the 6-DOF model, labeled d95.6. However, XR rig position data was not captured in this study so a similar 9-DOF model could not be created. Rather, a 9-DOF model in this study was created by using the combined eye gaze x (right/left), y (up/down), and z (forward) data from the Varjo eye tracking output.

Participants' data was broken up into one-minute epochs due to the interval that the FMS was asked. Epochs for participants who dropped out early were included if they completed the entire minute for a given epoch length. For example, a participant who dropped out 5.5 minutes into the exposure would still have valid epochs 1 - 5 but not epoch 6. This resulted in a total of 684 epochs for 6-DOF analysis and 671 for 9-DOF analysis. The difference is due to some epochs with invalid eye tracking but valid head tracking data.

The linear mixed models used to analyze the d95 scores were Equation 1 and Equation 2.

$$fms_score \sim condition * (d95.6 \text{ OR } d95.9) + (1 | ID) \quad (1)$$

$$fms_score \sim condition + d95.6 + d95.9 + (1 | ID) \quad (2)$$

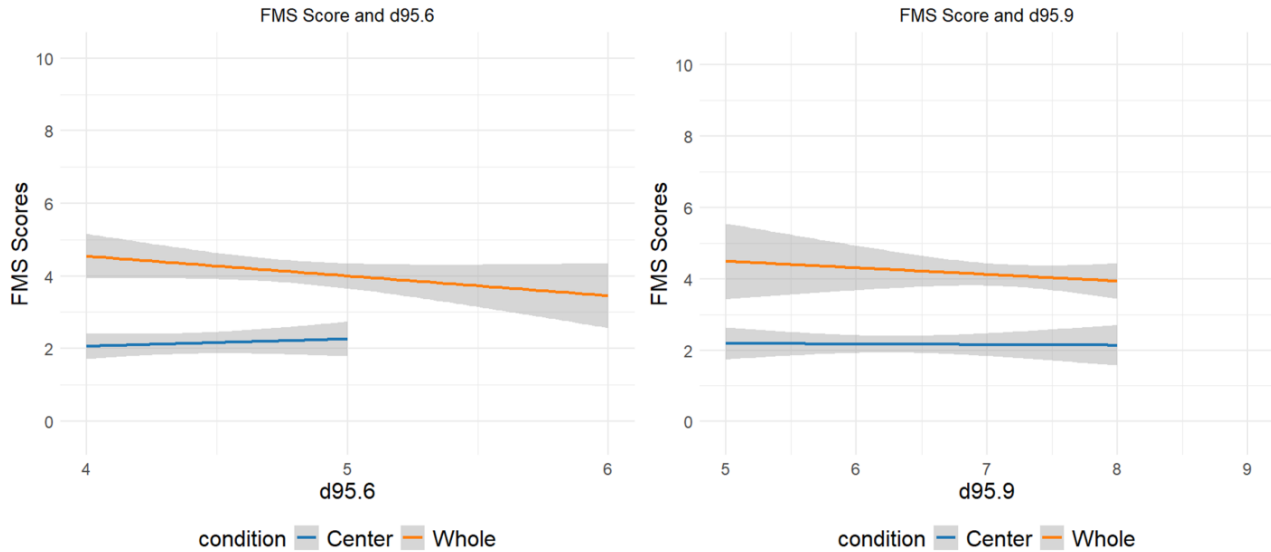


Figure 1: (Left) The impact of d95.6 on participants' FMS ratings across epochs. Each line represents a linear regression line for the two experimental conditions. The shaded grey area indicates the confidence interval. (Right). The impact of d95.9 on participants' FMS ratings across epochs.

The loadings were evaluated to determine how each principal component was constructed. For a principal component, the loading values were grouped into their modality – either head translation, head rotation, or eye movement. A total proportion of loadings for the three categories per epoch was calculated to determine the dominant modality. A threshold of 5% difference was set based on the generally accepted value for significance for categories that shared a similar proportion of loading, which would create a mixed category between the two similar loading modalities (e.g., translation + rotation). The resulting dominant loading categories were analyzed on a per-epoch basis and on a per-participant basis by counting the different categories in which each participant showed loading dominance. This process was completed for the first three principal components: PC1, PC2, and PC3. Statistical analyses were the goal of this effort; however, most of the resulting categories were so unbalanced that no statistical analyses could be performed.

4 RESULTS

4.1 d95 Scores

The linear mixed model (Equation 1) of the d95.6 score revealed significance for both the condition ($p < .01$, $b = 3.61$, $t(585.25) = 3.175$) and the d95.6 score ($p = .018$, $b = -0.29$, $t(624.47) = -2.382$). The interaction between condition and d95.6 was not significant ($p = 0.15$). These results are shown in Figure 1, left. The linear mixed model of the d95.9 score revealed similarly significant findings for both the condition ($p < .01$, $b = 4.75$, $t(659.76) = 3.112$) and the d95.9 score ($p = .033$, $b = -0.25$, $t(620.29) = -2.139$). The interaction between condition and d95.9 was not significant ($p = 0.07$). These results are shown in Figure 1, right. The linear mixed model of both the d95.6 and d95.9 scores together with the condition (Equation 2) revealed significance for only the condition ($p < .001$, $b = 2.18$, $t(79.64) = 4.57$). Neither d95.6 ($p = .21$) nor d95.9 ($p = .11$) were significant. This could be in part due to the degree of multicollinearity, with variance inflation factors (VIF) for d95.6 and d95.9 of VIF = 3.62 and VIF = 3.77 respectively. However,

since the VIF values are below 4, they do not require further investigation [20].

4.2 Principal Component Loadings

The first three principal components, PC1, PC2, and PC3, were analyzed individually. For PC1, the dominant modality was from the translation variables (x, y, z position), where 429 of the 671 epochs had the most variance caused by translation variables. For PC2, the dominant modality was from the eye variables (gaze x, y, z position), where 259 of the 671 epochs had the most variance caused by eye variables. Similar results occurred for PC3, where 441 of the 671 epochs had the most variance caused by eye variables. Rotation loadings were present in all components but were not often the highest contributor. Total variance explained by each principal component averaged across epochs can be seen in Figure 2.

Comparison of dominant modality distribution between early (1-5) and late (6-10) epochs revealed different patterns across the first three PCs. For PC1, modalities were collapsed into translation dominant and non-translation dominant due to small sample sizes, and a Fisher's exact test showed a significant reduction in translation dominant between early and late epochs ($p < .001$), but translation remained the dominant modality in both phases. However, based on chi-square tests no significant differences in dominant modality distributions existed between phases for PC2, $\chi^2(5) = 7.58$, $p = .18$, or PC3, $\chi^2(5) = 7.86$, $p = .16$.

To determine if the eye-dominant results of PC2 and PC3 overtook the translation dominance of PC1, total variance explained by each modality was summed across PC1-PC3 for every epoch. The dominant modality remained translation, with 531 epochs, followed by rotation with 101, and eye with 39.

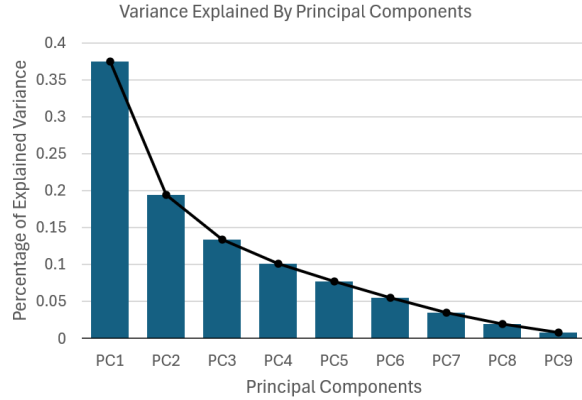


Figure 2: Percentage of average variance explained by each principal component across epochs for each PC in the 9-DOF data.

5 DISCUSSION

5.1 d95 Scores

Regarding RQ1, the results of this study were not significant for the d95.6 or d95.9 scores in the overall combined linear mixed model, despite both showing significance when analyzed individually. The only factor in the full model of significance was the experimental condition (Center vs. Whole). In this paradigm, the experimental condition appears to dominate variance in sickness ratings, suggesting that d95 plays a diminished role when motion is externally imposed. This result suggests that while the d95 captures motion complexity related to cybersickness under certain conditions, the predictive abilities may be sensitive to experimental contexts or modeling choices.

Regarding RQ2, including eye data into the d95 calculation produced similar results to the d95.6 model. Specifically, the independent analysis of d95.9 showed a significant negative relationship with FMS while the combined model did not. This indicates that while both d95.6 and d95.9 models behave similarly, no additional predictive power was gained by including eye gaze data.

Figure 1 illustrates the relationship between FMS and d95 for both experimental conditions. The relatively flat slopes, particularly for the Center condition, indicate that changes in motion complexity are not strongly associated with changes in FMS scores. The separation between the Center and Whole condition lines in both d95.6 and d95.9 illustrates that the condition has a substantial impact on FMS. Although the Whole condition has a negative slope, the magnitude is small due to the lack of significance of the combined model. While d95 is shown as continuous on the x-axis, it is a derived integer value and represents increasing motion complexity. Intermediate values (e.g., 5.5) should be interpreted as model-based interpolations rather than literal states.

This result differs from prior work where the d95 score was found to have a negative relationship with user FMS in the combined model. Several factors may explain the differing findings. Most importantly, the 9-DOF data in this study used eye-tracking variables in place of virtual camera rig position and rotation data used in the original study. The inclusion of eye tracking as a second behavioral subsystem did not increase predictive value of the d95 score, so although the data itself may be easy to capture during studies, it does not appear particularly valuable in the d95 context. These findings suggest that d95 use in

cybersickness may be specific to physical and virtual position and rotation modalities only, or to the level of user control or the task within the virtual environment. Additionally, the data in this study was collected from a virtual environment which contained only passive motion, where the user has limited agency over their motion. This context could limit generalizability of results. Future studies should investigate if these results hold when all the variables used for 9-DOF between this paper and the DeGuzman et. al. paper are used (physical head position and rotation, virtual head position and rotation, and eye gaze data).

Raw gaze data was used in this study, rather than derived eye-tracking metrics such as saccades or blink rate. This decision was motivated by the exploratory nature of the study to determine if unprocessed eye data can meaningfully influence the d95 metric. Future work should investigate if derived eye data metrics such as saccades, blink rate, and fixation stability have a meaningful effect on d95 scores over the raw gaze data.

5.2 Principal Component Loadings

The loading analysis provides insight into how the eye gaze data behaved as the newly included modality. PC1 was dominated by the translational variables, indicating that positional head movements accounted for the largest variance across most epochs. However, PC2 and PC3 were dominated by eye gaze variables. Rotational variables were present among each PC but never to a dominate level. When variance was summed across PC1-PC3, translation remained the dominant modality in a majority of epochs, suggesting that positional adjustments dominated the overall movement structure. Even though PC2 and PC3 were often eye-dominant, the combined variance across the first three principal components rarely exceeded the variance explained by translation alone.

Regarding RQ3, the loading analysis indicated that eye movements do contribute strongly to certain principal components, namely PC2 and PC3, but these components explain less overall variance than PC1, which is translation dominant. This suggests that even though eye movements are a separate and structured behavior, they do not dominate overall motion complexity. This helps explain why extending the d95 to eye data did not strengthen its predictive power over only head movement data alone.

Comparison between early and late phase dominant modalities showed that even though translation dominance frequency decreased from early to late for PC1, it remained as the most dominant modality. No significant changes in dominance existed for PC2 or PC3. These results suggest that the dominant modalities are relatively stable across the exposure period.

6 CONCLUSION

This study examined the generalizability of the d95 motion complexity metric in a cybersickness application by applying it to a new dataset and extending it to use eye tracking data. The results showed that the combined model did not show significance even while the independent models did, suggesting that the predictive power of d95 may be sensitive to the modeling context or to specific experimental environments. Including eye tracking data in the d95 did not improve the model's predictive power over head tracking alone, as d95.6 and d95.9 mirrored results of one another, suggesting that the observed limitations in this dataset may not be due to the inclusion of eye tracking data.

While DeGuzman et al. demonstrated that adding virtual rig motion increased the sensitivity of d95, their analysis did not examine how motion modalities contributed to the principal components. Our loading analysis extends this work by showing that d95 is dominated by head translation, even when other

behaviors (e.g., eye movements) are present. This analysis of principal component loadings gave insight into why eye tracking did not enhance predictive power. Eye movements were dominant for specific principal components, but the overall variance explained was still mainly dominated by translation variables. This indicates that while eye movements are structured and visible through PCA, they do not substantially contribute to the d95 behavior.

These findings help clarify the boundary conditions under which d95 is an objective indicator of cybersickness. d95 appears most informative when users actively control locomotion [3], and less sensitive when motion is passive, constraining behavioral degrees of freedom.

Together, the findings in this study indicate that d95 best reflects gross head movements rather than fine oculomotor movements. Future work should investigate how different tasks influence the relationship between motion complexity and cybersickness. Additionally, future work should ensure that physical head data, virtual head data, and eye tracking variables are all captured to validate the results of this study when virtual head data is included.

ACKNOWLEDGMENTS

This material is partially based upon work supported by the National Science Foundation under Award No. 2309990.

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